Attic: GPU CorrCal paper

Material removed from the main draft. Kept for the thesis and future papers.

Illustrative entries for the example project.

A1. FP32 accumulation of the Woodbury inner matrix

Cut: 2026-10-08
From: Methods, "Numerical precision"
Evidence: results.md 2026-10-07; `scripts/precision_test.py`
May resurface in: Thesis chapter on kernel design; possible mixed-precision follow-up

Tried single-precision accumulation in the batched inner-matrix build to halve memory traffic. For groups with many baselines the inner matrix $\mathbf{I} + \mathbf{\Delta}^\dagger \mathbf{N}^{-1} \mathbf{\Delta}$ became ill-conditioned enough that the FP32 Cholesky failed or gave a χ^2 differing from the CPU reference by [value]. Original section text:

> *To reduce bandwidth, the inner matrix is accumulated in single precision and factorised in double precision...*

Lesson for the main text: one sentence stating that double precision is required, with this entry as the source.

A2. Fused single-kernel likelihood and gradient

Cut: 2026-10-22
From: Methods, "Kernel design"
Evidence: results.md 2026-10-21 (Nsight Compute profile)
May resurface in: Thesis chapter on kernel design patterns

Attempted a single kernel computing the Woodbury solve, χ^2, and the gradient per group to avoid global-memory round trips. Register usage exceeded the per-thread limit and spilled to local memory; the fused kernel was slower than the split version by [factor]. Kept as a worked example of a fusion that does not pay off.

A3. Claim: CorrCal converges in fewer iterations than redundant calibration

Cut: 2026-10-29
From: Results, "Calibration accuracy"
Evidence: results.md 2026-10-28; iteration counts inconsistent across seeds
May resurface in: Nowhere unless new evidence; note for group meeting Q&A

An early run suggested faster convergence. Repeating over [N] random seeds showed no consistent difference, and the comparison depended on the choice of convergence threshold. Claim removed from the results claim box and the abstract.
